Online Shoppers Purchasing Intention Dataset Analysis Using Multivariate Techniques

Kalana Ekanayake
S/20/820

STA4053-Mini Project

Content

1 Introduction

1.1 Background

The digital marketplace has transformed the consumer behaviour moving from simple storefronts to more complex click-stream-based interactions. For e-commerce platforms, predicting whether a visitor completes a purchase is critical for personalized marketing and resource allocation. This study uses the “Online Purchase Intention” dataset to explore these behavioral patterns using multivariate analysis techniques.

1.2 Problem Statement

A high number of visitors to the website does not always mean high revenue. While many users browse through the products only a few will purchase products.The primary challenge is to identify which specific behaviours are associated with purchase intention of the consumers.

1.3 Objective

This study aims to :

- Reduce dimensionality - Use **Principal Component Analysis** to reduce the numerical metrics into smaller sets of behavioral dimensions.
- Evaluate relationships - Use **Canonical Correlation** to explore the relationship between user activity and site analytics.
- Predict purchase intention - Predict purchase intentions of the visitors from their session behavior Using **Discriminant Analysis.**

2 Methodology

2.1 Data Description

The dataset used in this study is designed for real time analysis of online shopper behavior, focusing on predicting purchasing intention. It consists of feature vectors derived from 12,330 sessions.The dataset was formed so that each session would belong to a different user in a 1-year period to avoid any tendency to a specific campaign, special day, user profile, or period. Of the 12,330 sessions in the dataset, 84.5% (10,422) were negative class samples that did not end with shopping, and the rest (1908) were positive class samples ending with shopping.

Feature Name	Feature Description	Min Value	Max Value	Mean	Standard deviation
Administrative	Number of pages visited by the visitor about account management	0	27	2.315	3.32
Administrative duration	Total amount of time (in seconds) spent by the visitor on account management related pages	0	3398	80.82	176.70
Informational	Number of pages visited by the visitor about Web site, communication and address information of the shopping site	0	24	0.5036	1.26
Informational duration	Total amount of time (in seconds) spent by the visitor on informational pages	0	2549	34.47	140.64
Product related	Number of pages visited by visitor about product related pages	0	705	31.73	44.45
Product related duration	Total amount of time (in seconds) spent by the visitor on product related pages	0	63973	1194.8	1912.25
Bounce rate	Average bounce rate value of the pages visited by the visitor	0	0.2	0.0221	0.04
Exit rate	Average exit rate value of the pages visited by the visitor	0	0.2	0.0430	0.05
Page value	Average page value of the pages visited by the visitor	0	361	5.889	18.55
Special day	Closeness of the site visiting time to a special day	0	1	0.0614	0.19

Table 1: *Numerical features used in the user behavior analysis*

Feature Name	Feature Description	Number of levels
Operating Systems	Operating system of visitor	8
Browser	Browser of the visitor	13
Region	Geographic region from which the session has been started by the visitor	9
Traffic Type	Traffic source by which the visitor has arrived at the web site (e.g., banner, SMS, direct)	20
Visitor Type	Visitor type as “New Visitor,” “Returning Visitor,” and “Other”	3
Weekend	Boolean value indicating whether the date of the visit is weekend	2
Month	Month value of the visit date	12
Revenue	Class label indicating whether the visit has been finalized with a transaction	2

Table 2: *Categorical features used in the user behavior analysis*

2.2 Data Preprocessing

Handling missing values:

No missing values were found in the dataset.

Handling Outliers:

Interquartile Range method flagged 7072 rows as potential outliers which represent 57.36% of the total dataset. Removing more than half of the data would compromise a significant amount of data, particularly by excluding valid, high intensity user sessions.Therefore no outliers were removed.

2.3 Data Visualization

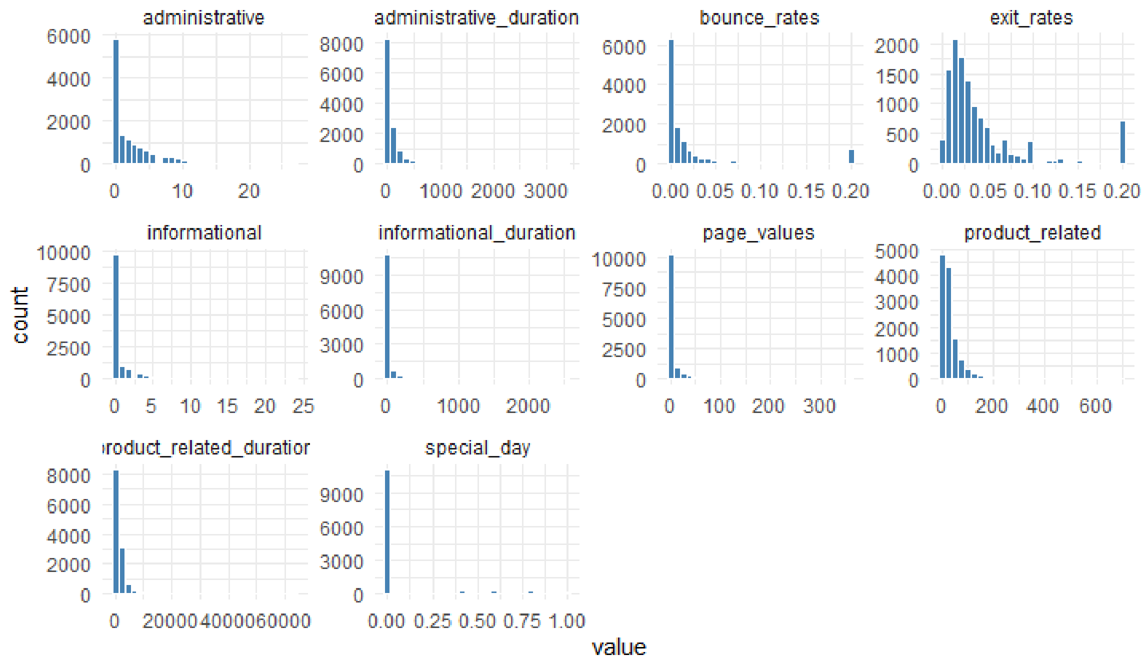

Figure 1: *Distribution of numerical variables*

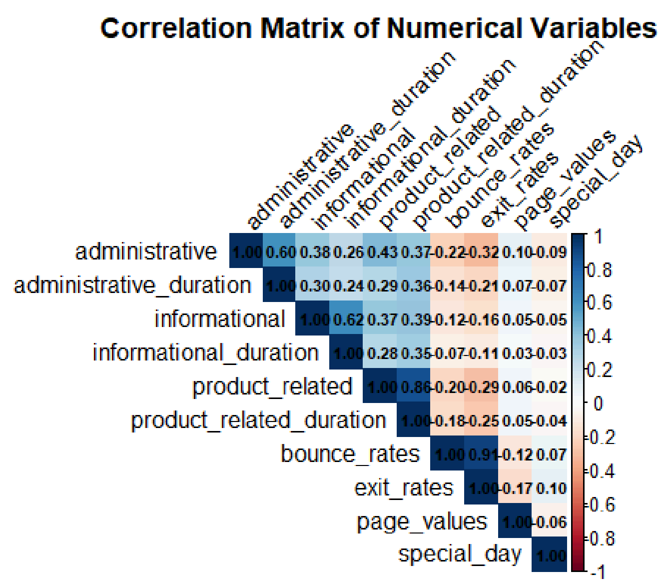

Figure 2: *Correlation of numerical variables*

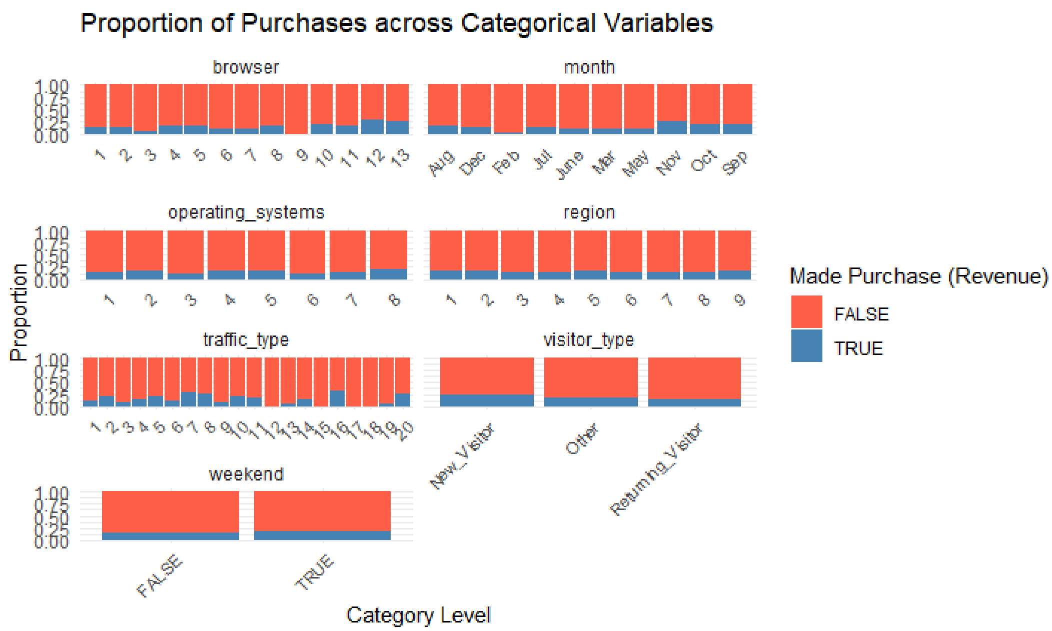

Figure 3: *Distribution of categorical variables*

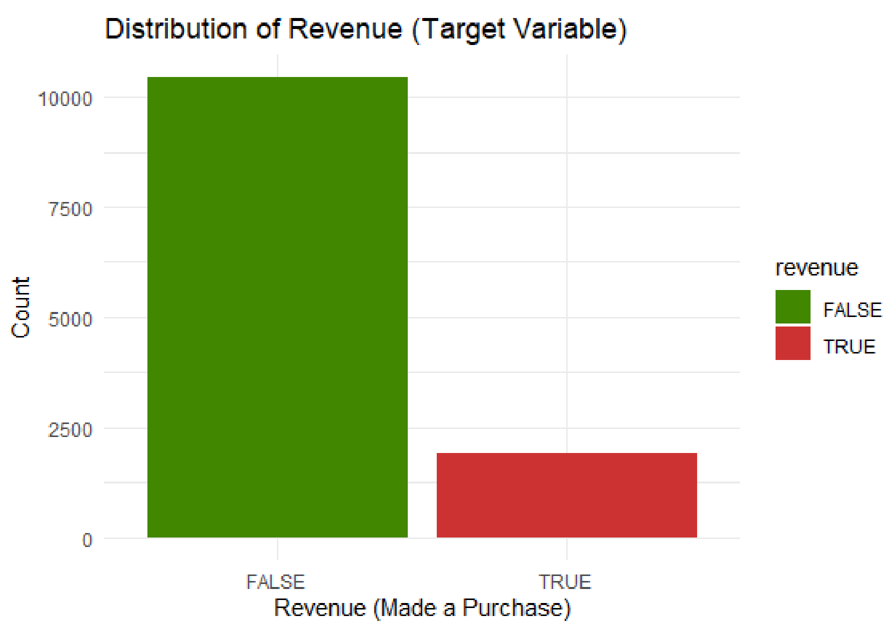

Figure 4: *Distribution of target variable*

2.4 Multivariate Methods

2.4.1 Principal Component Analysis (PCA)

Principal Component Analysis is a multivariate technique used for dimensionality reduction. In scenarios involving high dimensional datasets where many variables may provide redundant or overlapping information, PCA is used to transform the original data into a few, highly interpretable linear combinations.
The objectives of this analysis include,

- Dimension reduction
- Enhanced interpretability
- Explain as much of the total variance
- Handle Multicollinearity

In the case of the purchase intention dataset which include 10 numerical features, PCA was used to resolve multicollinearity between variables such as Administrative and Administrative duration by creating a smaller number of uncorrelated components.This dimensionality reduction eliminates redundant information and filters out noise, allowing the model to focus on the core patterns of shopper behaviour.

2.4.2 Canonical Correlation Analysis

Canonical Correlation Analysis is a multivariate method designed to investigate the complex relationships between two separate sets of variables measured on the same experimental unit. This method seeks to identify canonical variables from each set such that correlation between these combinations is maximized. These correlations are known as canonical correlations and they represent the strength between the two sets of variables.
The general objectives are,

- Dimension reduction - summarizing the dependencies between two sets of variables into few highly correlated canonical pairs.
- Variable contribution analysis - Identify which specific variables from each set contribute to the observed relationship

For the purchase intention dataset Canonical Correlation Analysis was used to check for relationship between variables associated with user activity (Administrative, Informational, Product Related) and set of variables corresponding to site analytic metrics (Bounce rate, Exit rate, Page values).

2.4.3 Discriminant Analysis

Discriminant Analysis is a multivariate classification technique used to categorize observations into predefined groups by identifying a function of predictor variables that maximizes the statistical separation between group means.
In this study the method was used to predict the binary target variable (Revenue) by distinguishing between visitors who finalized a purchase and those who did not. This technique helped uncover specific combinations of features that most effectively highlight the distinctions among the two types of users.

3 Results and Discussion

3.1 Principal Component Analysis Results

3.1.1 Selecting principal components

Principal Component	Eigenvalue	Proportion of Variance Explained	Cumulative proportion
PC1	3.4003 (>1)	0.34	0.34
PC2	1.6751 (>1)	0.1675	0.5076
PC3	1.0712 (>1)	0.1071	0.6147
PC4	1.0107 (>1)	0.1011	0.7158
PC5	0.941	0.09411	0.80987
PC6	0.9271	0.09271	0.90258
PC7	0.422	0.0422	0.9448
PC8	0.3516	0.03517	0.97995
PC9	0.1228	0.01229	0.99224
PC10	0.0776	0.00776	1

Table 3: *Eigenvalues, proportion of total variance and cumulative proportion for each PC*

Kaiser criteria suggests to retain the principal components whose eigenvalues are greater than 1. According to the above results only **four principal components** can be retained and these components account for 71.58% of the total variation.

3.1.2 Interpreting selected principal components

Variable	PC1	PC2	PC3	PC4
Administrative	**0.7050**	**0.0679**	**-0.2645**	**0.31101**
Administrative duration	**0.6069**	**0.1387**	**-0.3323**	**0.3665**
Informational	**0.6410**	**0.3647**	**-0.1564**	**-0.4746**
Informational duration	**0.5454**	**0.3945**	**-0.1462**	**0.6015**
Product related	**0.7588**	**0.1925**	**0.4088**	**0.2489**
Product related duration	**0.7624**	**0.2460**	**0.3753**	**0.2159**
Bounce rate	**-0.5135**	**0.7896**	**-0.1236**	**0.1995**
Exit rate	**-0.5986**	**0.7514**	**-0.0886**	**0.1424**
Page value	**0.1732**	**-0.2429**	**-0.3729**	**0.0286**
Special day	**-0.1309**	**0.1330**	**0.6120**	**-0.1519**

Table 4: *Loadings of selected PCs*

PC1: Explains 34% of the variance. Strongly positively correlated with product related, product related duration and administrative variables. This component represents the general engagement of the users. Higher values indicating users who spend a significant time browsing the site.

PC2: Explains 16.75% of the variance. Strongly positively correlated with the bounce rate and exit rate. This component represents the bounce behaviour of the user. Higher values identify sessions where users leave almost immediately without meaningful interactions.

PC3: Explains 10.71% of the variance. Dominated by special day and page value variables. This highlights the browsing behaviour of users specifically related to holiday seasons or special events.

PC4: Explains 10.11% of the variance. Negative correlation with informational duration, informational variables contrasted with administrative duration. This dimension separates users who seek for product information and those who focus on administrative tasks such as account management.

3.2 Canonical Correlation Analysis Results

Canonical correlation was used to investigate the multivariate relationship between two sets of numeric variables,

- Set1: User Actions which includes administrative, informational and product related page counts.
- Set2: Site Analytics which include bounce rate, exit rate and page values.

3.2.1 Canonical correlations and significance test

The analysis resulted in three pairs of canonical variates and likelihood ratio test using Wilk's lambda was done to check for significance

Variate Pair	Canonical Correlation	Wilk's lambda test p value
Pair 1	**0.4057**	**0.0000 (considering all pairs)**
Pair 2	**0.0367**	**0.0022 (considering 2nd and 3rd pair)**
Pair 3	**0.0015**	**0.8672 (considering only 3rd pair)**

Table 5: *Canonical correlations*

The first pair explains the majority of the relationship between the two sets while relationships explained by the other pairs are negligible. Although the first 2 pairs are statistically significant ($p<0.05$) only the first pair was considered due to the absence of meaningful relationships in the second pair.

3.2.2 Canonical coefficients for the first pair

Set 1 (Actions)	Coefficients	Set 2 (Analytics)	Coefficients
Administrative	-0.643	Bounce rate	-1.112
Informational	-0.010	Exit rate	1.898
Product related	0.531	Page values	-0.033

Table 6: *Canonical coefficients for the first pair*

User actions (U1): This variate is primarily driven by administrative and product related variables. This represents high intensity browsing while users are actively managing accounts and exploring products.

Analytics (V1): This variate is driven by the exit rate and bounce rate.

These sets have a moderate positive correlation (0.4057) between these sets. High engagement in administrative and product tasks are associated with decreased bounce rates.The results confirm that users who navigate multiple administrative or product pages have lower bounce rates (leaving after the first page). However as they eventually end the session they contribute to higher exit rates on those specific pages.

3.3 Discriminant Analysis Results

For the discriminant analysis the PC's that were selected from the previous section were used.

3.3.1 Checking for Assumptions

Since the principal components are uncorrelated with each other there is no multicollinearity present.
To assess the multivariate normality assumption a chi square QQ plot was used.

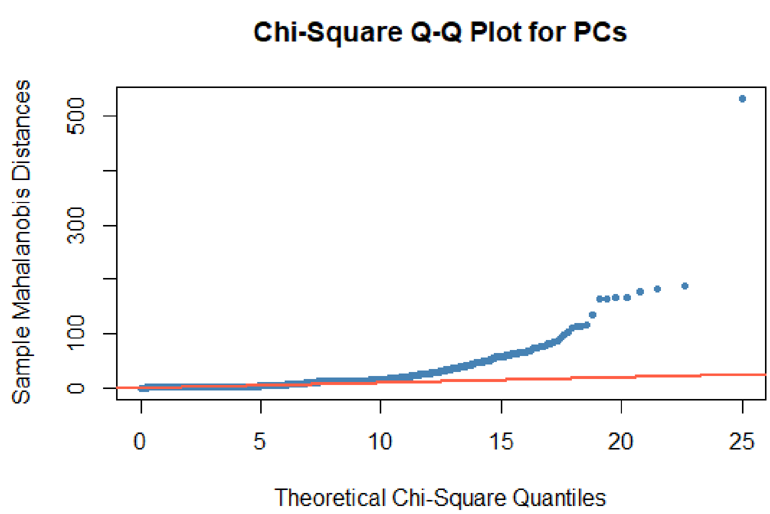

Figure 5: chi square QQ plot for normality check

According to the above plot the data seem to violate normality assumption. However with over 12000 observations discriminant analysis models will remain effective even if the data is not normal.
The Box's M test yielded a p value of 2.22e-16. Because this is highly significant (< 0.05) covariance matrices are heterogeneous. This violates a fundamental assumption of linear discriminant analysis. Thus quadratic discriminant analysis was implemented.

3.3.2 Quadratic Discriminant Analysis

The dataset was partitioned into a training set (80%) and test set (20%) to evaluate on unseen observations. Prior probabilities were estimated from the training data which reflects the actual class imbalance 84.52% for no revenue (False) and 15.48% for revenue (True). A QDA model was then fitted on the training data.

The resulted misclassification probabilities are,

		Predicted	
		False	**True**
Actual	**False**	0.9231	0.0768
	True	0.7373	0.2626

Overall cross validated misclassification probability: 17.91%

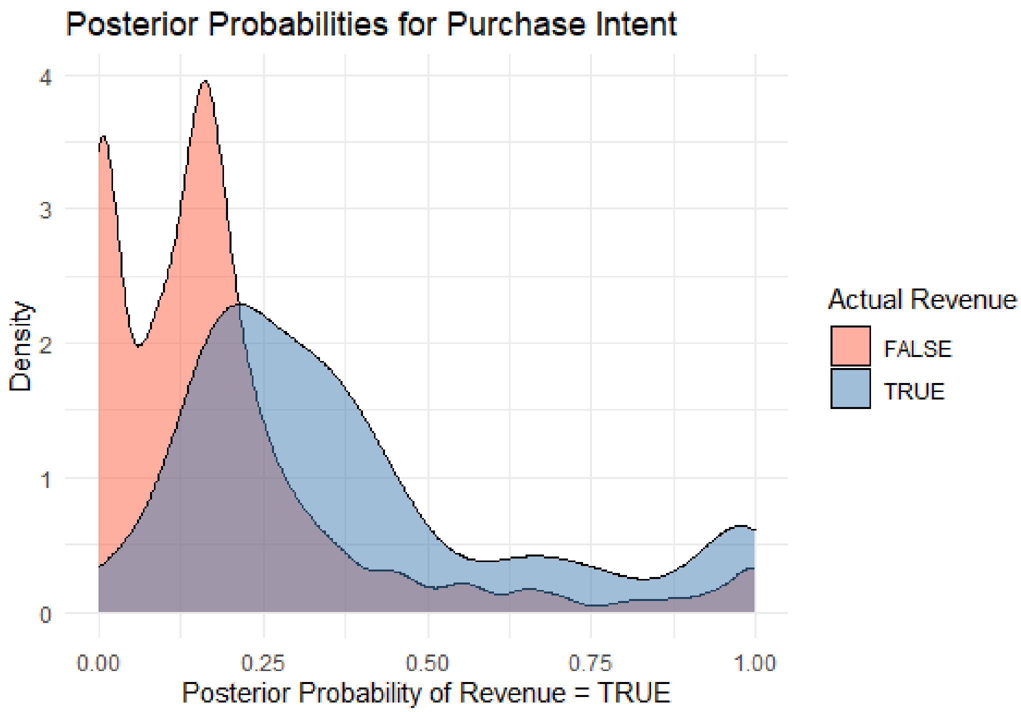

Figure 6: Posterior probabilities for purchase intent

The model was then evaluated on test data yielding an accuracy of 81.38%.
Overall the QDA model successfully handles the heterogeneous variance structure found in the dataset but is heavily influenced by the class imbalance. While the overall accuracy is high, the high misclassification rate for buyers suggests that browsing patterns for purchasers and non purchasers are quite similar.

3.4 Limitations

PCA and CCA all assume linear relationships between variables. Complex nonlinear patterns or interactions may not be captured, possibly oversimplifying the true structure of the data. Techniques such as QDA assume that the data follow a multivariate normal distribution within each group. If this assumption is violated (as is common in real-world datasets), the results might be biased or less reliable.

4 Conclusion

The study successfully applied multivariate statistical techniques to analyze and predict online purchase intentions. Through Principal component analysis (PCA) ten numerical features were reduced to four primary behavioural dimensions which accounted for 71.58% of the total variance.
The canonical correlation analysis revealed a significant relationship between user actions and site analytics. Specifically, high intensity browsing was associated with decreased bounce rates though it led to higher exit rates as sessions concluded.

Finally the quadratic discriminant analysis model achieved an overall accuracy of 81.38% in predicting purchase intent. However the model's performance was heavily influenced by class imbalance towards non purchasers.

5 References

- Sakar, C. & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5F88Q.
- Maćkiewicz, A., & Ratajczak, W. (1993). Principal components analysis (PCA). Computers & Geosciences, 19(3), 303–342. https://doi.org/10.1016/0098-3004(93)90090-r
- Wu, R., & Hao, N. (2022). Quadratic discriminant analysis by projection. Journal of Multivariate Analysis, 190, 104987. https://doi.org/10.1016/j.jmva.2022.104987

AI Disclosure

In accordance with the guideline for this project, I declare the use of generative AI (Gemini) for the purpose of grammar correction, rephrasing, code error correction and structural organization of the report.

6 Appendices

Complete R code

 # Loading Libraries

```
library(dplyr)
library(janitor)
library(factoextra)
library(corrplot)
library(ggplot2)
library(tidyverse)
library(CCA)
library(heplots)
library(MASS)
library(caret)
library(CCP)

set.seed(820)
```

 # Loading dataset

```
dataset <- read.csv("./online_shoppers_intention.csv")

dataset <- clean_names(dataset)

dataset <- dataset %>% mutate(across(c(operating_systems, browser, region, traffic_type,
visitor_type, weekend, revenue, month), as.factor))
dataset <- dataset %>% mutate(across(c(administrative, administrative_duration, informational,
informational_duration,
                                product_related, product_related_duration, bounce_rates, exit_rates,
page_values, special_day), as.numeric))

categorical_variables <- colnames(dataset %>% dplyr::select(where(is.factor)))
numerical_variables <- colnames(dataset %>% dplyr::select(where(is.numeric)))

numerical_data <- dataset %>% dplyr::select(all_of(numerical_variables))
categorical_data <- dataset %>% dplyr::select(all_of(categorical_variables))
```

```r
str(dataset)

summary(dataset)


# Data preprocessing and visualization

cat("Number Of rows with missing values:", sum(dataset %>% is.na()))

numerical_data %>% pivot_longer(cols = numerical_variables) %>%
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ name, scales = "free", ncol=4) +
  theme_minimal()


p <- categorical_data %>%
  pivot_longer(cols = -revenue, names_to = "variable", values_to = "value") %>%
  ggplot(aes(x = value, fill = revenue)) +
  geom_bar(position = "fill") +
  facet_wrap(~ variable, scales = "free_x", ncol=2) +
  theme_minimal() +
  labs(title = "Proportion of Purchases across Categorical Variables",
       x = "Category Level",
       y = "Proportion",
       fill = "Made Purchase (Revenue)") +
  scale_fill_manual(values = c("tomato", "steelblue")) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size=8))
p

cor_matrix <- cor(numerical_data)

corrplot(cor_matrix, method = "color", type = "upper",
         tl.col = "black", tl.srt = 45,
         addCoef.col = "black", number.cex = 0.7,
         title = "Correlation Matrix of Numerical Variables", mar=c(0,0,1,0))

numeric_vars_long <- scale(numerical_data) %>%
  as.data.frame() %>%
```

```
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")

ggplot(numeric_vars_long, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot(outlier.colour = "red", outlier.shape = 8) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none") +
  labs(title = "Boxplots of Standardized Numerical Variables",
       y = "Standardized Value", x = "Variable")

is_outlier_iqr <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR_value <- Q3 - Q1

  lower_bound <- Q1 - 1.5 * IQR_value
  upper_bound <- Q3 + 1.5 * IQR_value

  return(x < lower_bound | x > upper_bound)
}

outlier_matrix <- sapply(numerical_data, is_outlier_iqr)

rows_with_any_outlier <- rowSums(outlier_matrix) > 0

total_rows <- nrow(dataset)
outlier_rows <- sum(rows_with_any_outlier)
percentage_removed <- (outlier_rows / total_rows) * 100

cat("Total rows originally:", total_rows, "\n")
cat("Rows flagged for removal:", outlier_rows, "\n")
cat(sprintf("Percentage of data that would be removed: %.2f%%\n", percentage_removed))

ggplot(dataset, aes(x = revenue, fill = revenue)) +
  geom_bar() +
  theme_minimal() +
  labs(title = "Distribution of Revenue (Target Variable)",
       x = "Revenue (Made a Purchase)", y = "Count") +
  scale_fill_manual(values = c("chartreuse4", "brown3"))
```

 # PCA Analysis

```
numerical_data <- dataset %>% dplyr::select(all_of(numerical_variables))

pca_result <- prcomp(numerical_data, scale = TRUE)


print("Mean:")
pca_result$center
print("Variance:")
pca_result$scale

summary(pca_result)
# According to Kaizer criteria choose 4 PCs
variances <- pca_result$sdev^2
print(variances)

pca_result$rotation

fviz_eig(pca_result, addlabels = TRUE, ylim = c(0, 50))

var <- get_pca_var(pca_result)
loadings <- var$coord
print(loadings)
```

 # Canonical Correlation

```
scaled_data <- as.data.frame(scale(numerical_data))


set1 <- scaled_data %>% dplyr::select(administrative, informational, product_related)

set2 <- scaled_data %>% dplyr::select(bounce_rates, exit_rates, page_values)

cca_result <- cc(set1, set2)

cat("Canonical Correlations:\n")
print(cca_result$cor)

cat("Coefficients for Set 1 (Actions):\n")
print(cca_result$xcoef)
```

```
cat("Coefficients for Set 2 (Analytics):\n")
print(cca_result$ycoef)

plt.cc(cca_result, var.label = TRUE)

rho <- cca_result$cor
n <- nrow(dataset)
p <- ncol(set1)
q <- ncol(set2)
p.asym(rho, n, p, q, tstat = "Wilks")

d_data <- as.data.frame(pca_result$x[, 1:4])
d_data$revenue <- dataset$revenue


train_idx <- createDataPartition(d_data$revenue, p = 0.8, list = FALSE)
train <- d_data[train_idx, ]
test <- d_data[-train_idx, ]


box_m_pc <- boxM(train[, 1:4], train$revenue)
print(box_m_pc)


mu <- colMeans(train[, 1:4])
sigma <- var(train[, 1:4])
d2 <- mahalanobis(train[, 1:4], center = mu, cov = sigma)

qqplot(qchisq(ppoints(nrow(train)), df = 4), d2,
    main = "Chi-Square Q-Q Plot for PCs",
    xlab = "Theoretical Chi-Square Quantiles",
    ylab = "Sample Mahalanobis Distances",
    pch = 20, col = "steelblue")
abline(0, 1, col = "tomato", lwd = 2)


qda_model <- qda(revenue ~ ., data = train)

qda_cv <- qda(revenue ~ ., data = train, CV = TRUE)
conf_matrix_cv <- table(Truth = train$revenue, Classified = qda_cv$class)
```

```
print(prop.table(conf_matrix_cv, margin = 1))

overall_error_pc <- 1 - (sum(diag(conf_matrix_cv)) / sum(conf_matrix_cv))
cat(sprintf("Overall CV Misclassification Probability: %.2f%%\n", overall_error_pc * 100))

qda_preds <- predict(qda_model, newdata = test)
final_acc_pc <- sum(diag(table(test$revenue, qda_preds$class))) / nrow(test)
cat(sprintf("\nFinal PCDA Accuracy: %.2f%%\n", final_acc_pc * 100))

plot_data <- data.frame(
  Prob_TRUE = qda_preds$posterior[, "TRUE"],
  Actual_Revenue = test$revenue
)

ggplot(plot_data, aes(x = Prob_TRUE, fill = Actual_Revenue)) +
  geom_density(alpha = 0.5) +
  theme_minimal() +
  labs(
    title = "Posterior Probabilities for Purchase Intent",
    x = "Posterior Probability of Revenue = TRUE",
    y = "Density",
    fill = "Actual Revenue"
  ) +
  scale_fill_manual(values = c("tomato", "steelblue"))
```